

Collective Behavior (Lecture) ~ Daniel Friedman ~ Active Inference for the

Courses/ActiveInferenceForTheSocialSciences | Daniel Friedman ~ Active Inference for the Social Sciences 2023 | 1:17:31

Hello, everyone. It is August 2nd, 2023. We're in the Active Inference for the Social Sciences course. I'm Daniel Friedman, and this is going to be a lecture on collective behavior. Now, of course, disclaimers could abound, and there's so many ways to go about this topic of collective behavior that in actuality, in enactment, I really hope it is multi-scale and collective.

I'm really looking forward to your comments and your engagement with this area, with this lecture, with this course. So if you're watching live, you can add comments into the live chat or questions, and I'll check them out at the end. Also in the YouTube live chat and in the video description, there is a link to where you can submit questions for all of the teachers of this course.

That's going to be a coda form. And just more generally, let's have fun enacting ecosystems of shared intelligence together. And I hope that this lecture can be something like one attempt to make that happen epistemically. We're going to have many fun things to talk about. Collective behavior, what differentiates it from simply behavior? I mean, aren't all behaving systems assemblages? Is not the brain composed of multiple parts, brain and body, multiple people? What are we really talking about with individual behavior? Why do we need adjectives at all for behavior? What is a collective and how would we even know it except through its behavior? So there's going to be all kinds of fun topics. Avel provided me with this image and suggested that it would be a great place to head towards. And so we can think of this as where we're going. And where we're going is understanding the agents or the human in a social relationship through social action and the recipient of social constraints and observations. Through that interface or blanket or holographic screen, the agent or the human is engaging on this sociocultural landscape. And so we can think about that agent of engagement with a social world and the dynamic of sociocultural constraints. Now that's where we're going and that's how someone else has framed where we're going. What I'll tell you here is how we're getting there. How we're getting there, it's going to be a lot of fun. There's going to be a lot of ants and the door is going to be wide open. So everyone is a nest mate on this slide and we're going to be with our regimes of attention and our stigmatic trails just absolutely up, down, left, right, all over this sociocultural landscape. All right. So here's the plan for the collective behavior lecture. It's going to be presented as a sequence of views. Each view or perspective is going to consist of at very least a central citation and then one or more reflections on that citation. Four views on collective behavior taken as a bundle will constitute a main section of this lecture. And there are going to be three main sections of this lecture split on one hand into three groups of four and on the other hand into two groups of six. Splitting into fours, the main three sections are going to be background and context, which is going to provide some views and insights that are from outside of active inference because surprise, surprise, what was true before active inference or outside of active inference is also true within and after. Then we're going to turn to active inference specifically. We're going to talk about active inference insights and fundamentals for

collective behavior and applications and implications of utilizing or of conceptualizing of a social system as an active inference system. So those second two sections are each going to center around also specific recent work in the field. To split it into two groups of six, the first six views that are presented, so all of background and context and then half of the insights and fundamentals are going to be in support linguistically or rhetorically of a declarative statement. Whereas the latter six views are going to support or open up to a question. And that'll lead us to the final section, the fourth section, which is some open questions. At that time, I will look at the live chat and review any questions or thoughts that people have. And I'll also just open the door and leave it open with asking questions because that's something that we're going to be engaged in for so, so long. So let's go right into the first section, background and context. Here are the four views and the four declarative statements. Big picture. Many, if not all of the concepts, discussions, questions, and views on collective behavior from before or outside of active inference also apply after or inside active inference or in the active inference ecosystem. And so to support and to build out that point, I'm going to talk about four different works and some takeaways. First, watching behavior is complex. Here we'll look at Deborah Gordon's 1992 paper, Wickenstein and Ant Watching. Second, all intelligence is collective intelligence, which was a presentation of the same name by J.F. Cloutier earlier this year, referencing a paper of the same name. Third, We Have Never Been Individuals, a work by Gilbert et al. from 2012, A Symbiotic View of Life, We Have Never Been Individuals, a little bit of the play on We Have Never Been Modern. And the metaphors and frameworks and language that we use to talk about behavior, to talk about collective behavior, what adjectives we put in front of what nouns, that matters. And a great work that Ali shared with me that I think makes this point super well is by Navarro and Gabrielitis from 23. And that work is the morphology of collective intelligence, network, swarm, multitude, or assemblage. So let's head straight to Professor Gordon's 1992 work on Wickenstein and ant watching. And so just also for context, Professor Gordon was my PhD advisor. And so I was able to engage with this material and perspective through asynchronous means, such as this paper, but also seeing it actually play out on a sociocultural landscape in the desert in Southeast Arizona. That's where the complexity really flowered. And so research in animal behavior begins by identifying what animals are doing. In the course of observation, the observer comes to see animals as performing a particular activity. How does this process work? How can we be certain that behavior is identified correctly? Wickenstein offers an approach to these questions, looking at the uses of certainty rather than attempting to find rules that guarantee it. Here, two stages in research are distinguished. First, watching animals. And second, reporting the results to other scientists. Certainty about what animals are doing has different uses at each stage. And this is really a beautiful and subtle point. By distinguishing stages of the ethological encounter, the behavioral scientific encounter, whether we're talking about ants on the ground or people in the cities, separating out the observational moments and the uncertainties that are always at play there from this secondary communication about behavior, at which point having really clear, defined articulations of behavior

becomes so important for scientific reproducibility, integrity, accessibility, and so on.

Personally, I find that this paper anticipates a lot of the specifications that today we would consider part of the Meta-Bayesian perspective, which is to say we're making inferences as cognitive systems about other cognitive systems. She wrote,

My questions about ant-watching do not arise when I'm watching ants. Most of the time, I do not think much about certainty when ant-watching.

Replace that for the social human setting with people watching. Sometimes we're just watching people and that's okay.

That's part of the moment. I do not think much about certainty when ant-watching. I enact it.

The questions do not come from doubts about the certainty of my seeing, but from the ways that doubts about the certainty of seeing have shaped scientific practice.

The resolution of the problem would come from changes that would allow observers to talk about what they are actually doing when watching animals.

Truthful accounts of scientific animal watching don't yet exist. This truth is not yet formed because we have not yet learned to talk about it.

What a wonderful expression. And delightfully, I came to see the confused foragers as patrollers searching for new food.

I saw the other foragers as already decided about their destination, not available to be distracted by a new source of food.

And so we have this really fascinating dialectic where in the moments we're observing in this pre-definitional space, not questioning,

though maybe we went to the field site to ask and answer a question, get data about a question.

But in the moment we're in the flow of observation and not asking questions, though the uncertainty may be the highest.

Afterwards, when we're communicating to our scientific colleagues, I defined an outgoing forager as a forager with as an as an estimate with no object in the mandibles and heading off in a straight line, passing a demarcated point three meters from the colony entrance.

Then people, why three meters? Why not two meters? Or why can't it be carrying something on the way out? Those questions arise in that precise space of communication.

And I wanted to open the lecture with this view because it puts a space between I see that person putting out a fire.

They're a firefighter. In that space between is the interpretive moment and possibility.

And reimagining the social has so much to do with this process of, let's just say, ant watching.

Second view in this background and context section.

And here I wanted to draw from a wonderful presentation that J.F. Cloudier, a facilitator and participant at the Active Inference Institute, presented as part of his robotics and embodied project.

Where yes, he's building Lego robots that do symbolic active inference, but also opening up some great philosophical directions and ones that have a lot of implications for how we think about natural, artificial and synthetic intelligence.

This presentation is called all intelligence is collective intelligence.

And it was based upon the title and drawing upon a recent 2022 citation by Fallon days at all.

And some of these maxims or expressions that J.F. provided, I thought were so provocative and also brought us back to that initial question.

Why collective behavior? I mean, what system isn't a collective?

So why do we need the adjective if we're talking about behaving systems?

What is individual behavior? Individual what? Oh, just one bird. Oh, just one person. Okay. Made up of what?

And so J.F. wrote some expressions and you can see from the link and look a little more with some images and more of what he wrote.

But a few of these I thought were just excellent. Intelligence is a process enacted by interacting parts, not a property of individuals.

Again, whatever that would even mean. Inaction is death. Being by doing. Action without prediction is reckless.

No prediction without sense making. No sense made without a self versus the world. Or a self in the world.

A collective is differently intelligent than its members. Maybe trivially different. Maybe better in every way. Maybe another direction in every way.

Why do individuals participate in collectives? Why are people in groups?

Because sacrificing the autonomy to the collective provides benefit, survival or even thrival benefits.

That's that I to we. No making of sense without the self acting in or on the world. It's not just some essence that's disembodied. This is what embodiment is all about.

An autonomous agent is a self organizing collective of parts. Parts of a collective can themselves be autonomous agents. Collectives form to improve the chance of survival of their members.

And an agent in a collective benefits from the intelligence of the collective.

So, to me, this presentation and line of research pursued by JF and others absolutely puts a bow on the what is an individual, what is a collective, why do we even talk about, quote, individuals, or, quote, collectives, when we recognize the multi-scale nature of complex adaptive systems.

Third perspective.

A symbiotic view of life. We have never been individuals. This is a 2012 paper by Gilbert, Sapp, and Tauber.

And just to read a few aspects of what they wrote in the abstract.

The notion of the biological individual is crucial to studies of genetics, immunology, evolution, develop, anatomy, and physiology.

Each of these biological sub-disciplines has a specific conception of individuality, which has historically provided conceptual context for integrating newly acquired data.

Few descriptions.

Animals cannot be considered individuals by anatomical or physiological criteria, because a diversity of symbionts are both present and functional in completing metabolic pathways and serving other physiological functions.

Similarly, these new studies have shown that animal development is incomplete without symbionts.

Symbionts also constitute a second mode of genetic inheritance, providing selectable genetic variation for natural selection, which connects it to the evolutionary concept of an individual.

So if we pursue the metabolic.

So if we pursue the metabolic.

Oh, it's all the metabolic networks and how they're all connected.

That's the individual.

All right.

Well, that includes the microbiome and symbionts.

Well, no, it's the physiology and the decision-making and the neurophysiology.

Those are also in play with the symbionts.

Well, at least evolution only selects upon the germline, right?

Well, no, actually, there are these multiple mechanisms of inheritance, whether they're relevant in a given situation or not is situational.

But in principle, they exist and they're increasingly coming to be recognized.

Additionally, for the immune system, commonly described as, quote, self versus non self or self identification or self other discrimination.

After all, we talk about auto grafting and allo grafting, but of course, there are aspects of, quote, self or self under some definitions, like an epitope from a protein in cartilage, that an autoimmune disorder might target.

So the immune system is mobilized against self under some definitions, sometime, and is not mobilized against other under some definitions, sometimes.

Like we're not constantly mounting a massive immune response to our gut microbiome or skin microbiome.

So, recognizing the holobiont, the multicellular eukaryote, plus its colonies of persistent symbionts, as a critically important unit of anatomy, development, physiology, immunology, and evolution, and cognition, opens up a new investigative avenue and conceptually challenges the ways in which the biological subdisciplines have heretofore characterized living entities.

Here we have the cover of the famous work, Leviathan.

It's a large person constituted of other people.

Really making the point clear, but if we zoomed in on, quote, an individual person, wouldn't we find something like societies of cells?

What is an individual?

When is an individual what?

And what's on the line?

I thought, those are such big questions.

Maybe could we connect it to a little bit of more of a bibliographic piece of data just to show a snapshot and to open up this discussion to you all about what's on the line with what's an individual?

And so what I did is I searched for some literature trends on PubMed.

And here is the results per 100,000 citations for collective behavior.

And so we see collective behavior first being mentioned in the 1950s, at least in this literature database, and rising up on through around 500 or 400 or 500 per 100,000.

So 1,000 out of 100,000 would be 1%.

So half of a percent of all the citations on a major bioliterature database have collective behavior in them.

So that's a huge amount of work, and it looks like it goes down.

But when you actually look at how many citations there are, it's been about 4,000 to 5,000 publications per year.

It's just that the rest of the literature database may be expanding at a faster rate.

So thousands of papers per year and an appreciable fraction of work has to do with collective behavior.

And so I thought, wow, that's exciting.

That's great.

Well, how are we doing compared to behavior?

Now, collective behavior is this lower line in red.

And yeah, there's a blip, but behavior is up between 15 and 20% of citations in PubMed.

Now, maybe behavior is also referring to the behavior of a given material that may or may not be a superconductor or might be using different understandings of behavior.

But what can we say and what can we know about the difference between individual and collective behavior?

Now, interestingly, here I searched for individual behavior as a specific term.

So we see a lot lower, but individual behavior.

I mean, can the tide have already turned?

Can it really be the case that after 2020, this is a real meaningful decline?

I don't know.

It's too early to say.

We can't do technical analysis just like that, but very provocative and gives us some ways to look at the broader scale or collective scientific cognitive process associated with these incredibly important philosophical and biological questions about individuality.

Because, of course, to distinguish collective behavior from individual behavior or why would you even need an adjective at all?

It's going to come down to what is an individual and what's on the line so much.

And the last view here.

Drawing upon the work Morphology of Collective Intelligence, Network, Swarm, Multitude, or Assemblage.

And Tony Devaro and Alejandra Lopez-Gabralides wrote,

Network, Swarm, and Multitude are some of the different metaphors that have been used to understand the concept of collective intelligence.

All of them have reflected our relationship with the internet and its development.

With the advent of artificial intelligence, AI, we wonder about other metaphors that might offer a new understanding of collective intelligence to address today's environmental and sociotechnical challenges.

They bring up Network, Swarm, and Multitude to advocate for the fourth option, assemblage.

Understanding collective intelligence as an assemblage allows for the inclusion of different human and technological agents and overcomes the antipocentrism, human centrism, human exclusionism, of other models based on human will,

whether extended as in the multitude or suppressed as in the swarm.

Connotations here.

Connotations here.

design, influence, and implementation tools. And here I'm going to draw on two pieces of work,

epistemic communities under active inference by Mao and colleagues in 2022. And just from a several days ago, the work by Connor Hines and colleagues, collective behavior from surprise minimization. We'll unpack the point more when we get there. Second, collectives can be understood of consisting as consisting of action perception loops, which subsume, entail, or supervene upon individual level action perception loops. Niche modification, stigmagy, can also be modeled this way.

We'll draw upon the paper active inference, an active inference framework for ant colony behavior by myself and colleagues in 2021. And then we'll move into the questions and things will start to open up. So if it feels like we've been climbing uphill, we're two thirds of the way with the uphill climb. It feels like we've been going downhill. We're two thirds of the way with the downhill.

How do the concepts of generalized synchrony and homeostasis, which are so core to active inference, relate to collective behavior? Here we'll discuss the work by Raph Kaufman et al. from 2021, an active inference model of collective intelligence. And lastly, in this section, how can active inference be used to explore the balance between individual and collective behaviors and the bi-directional influence of collectives on individuals? And of course, vice versa. Here, we'll draw upon the computational boundary of a self. Developmental bioelectricity drives multicellularity and scale-free cognition by Mike Levin in 2019. So let's get right into the work by Mao and Connor and colleagues. First, I'm just going to bring this paper up. However, first, you can watch more about it in guest stream 17.1 from March, 2022. Here's two, two collectives right here. And also Mao will be covering this work in a lot more detail later on in the course. I just wanted to bring it up because I think it's one of the exceptional examples that can be deployed to show like, Hey, just thinking about the social as an active inference system is awesome. It's an intuition pump. It brings us into the active inference ecosystem. But then what do we do when we're there? And what we do when we're there is we develop generative models that actually take us towards where we want to be. Again, framing a situation as being related to active inference, or you could do an active inference model of that does very, very little work by itself. However, making generative models of situations, even if they're speculative generative models or qualitative or hypothetical, that can start getting towards doing work. But if someone said, well, that health condition, you can do a linear regression. Okay, but which treatment do we do? Well, it could be done with a linear regression. We could use a linear regression to figure out what method to use. Okay, but you know, more or less, hotter room or colder room, or just use a linear regression. So that's the conversation that we're having a lot with active inference. A lot of times people are talking about whether a given situation can or cannot be modeled with active inference or just in the abstract, what does active inference say about this situation? And again, the framework itself is general to the point of doing little work, and we love it that way. It enables us to move into systems of interest and deploy generative models. So what they did in this paper was leverage the active inference framework to provide an in-silico model on the computer chip model of confirmation bias and its effect on echo chamber formation. They had a system of interest, people's social engagement on so-called social media, and they had even a phenomena of interest, which was echo chamber formation.

They built a model based on active inference where agents tend to sample information in order to justify their own view of reality, something we call self-evidencing. We show that once agents have reached a certain level of certainty about their beliefs, it becomes very difficult to get them to change their views. The system of self-confirming beliefs is upheld and reinforced by the evolving relationship between an agent's beliefs and observations, which will, over time, will continue to provide evidence for their ingrained ideas about the world. Well, where I'm from, we drive on the right side of the road. And so you drive on the right side of the road, I drive on the right side of the road, we're not going to have an issue. Everywhere I look, I see cars moving around, I see people passing on the right on the sidewalk. It's all making sense. We're bounding surprise, we're doing active inference, surprise minimization, goals, reward, never need to come into play to give an account of how those systems could arise and be stabilized. These epistemic communities are consolidated by those shared beliefs and in turn tend to produce perceptions of reality that reinforce those shared beliefs. And on the top right is just a little teaser of what a Bayesian graphical model looks like of the kind that they deployed in this paper. And here we see some of the results from their parameter sweeping, exploring different combinations of parameters and looking at topics like the polarization index and re-attendance rate. And so all kinds of statistics that can be drawn upon connected to the generative model and also all kinds of opportunities and interfaces to relate this to empirical data as well as suggest experiments. So Mao is going to talk more about this and it's awesome work. And then late breaking news. Only several days ago, five or so days ago, an amazing new preprint was put up by Connor Hines and colleagues. And it was just too cool not to include here and reflects the very rapid pace of collaborative development in the active inference ecosystem. And this is their work, collective behavior from surprise minimization. And they made every point I wanted to make here and now. Collective motion is ubiquitous in nature. Typically such macroscopic patterns arise from decentralized local interactions among constituent components, e.g. individual fish in a school. There's that word individual again. Preeminent models of this process describe individuals as self-propelled particles subject to self-generated motion and social forces such as short-range repulsion and long-range attraction or alignment. However, organisms are not particles. They are probabilistic decision makers. They're cognitive particles. We introduce an approach to modeling collective behavior based on active inference. This cognitive framework casts behavior as the consequence of a single imperative to minimize surprise. Again, if you want to, you can contrast that with reward learning where the imperative is to maximize reward. In active inference and the free energy principle, we're taking a path of least action, not the laziest path or the most passive one, but a little bit more technically a path of least action. And we're minimizing surprise. We're bounding surprise with variational free energy. And it turns out that that helps us minimize surprise and maximize model evidence. And it can include things like novelty and curiosity seeking very naturally. So this doesn't mean we're squeezing the juice out of life at all. We demonstrate that many empirically observe collective phenomena, including cohesion, milling, and directed motion emerge naturally when considering behavior as

driven by active Bayesian inference without explicitly building behavioral rules or goals into individual agents. No reward. No goal. No master of the collective. Furthermore, we show that active inference can recover and generalize the classical notion of social forces as agents attempt to suppress prediction errors that conflict with their expectations. By exploring the parameter space of the belief-based model, we reveal non-trivial relationships between the individual beliefs and group properties like polarization and the tendency to visit different collective states. We also explore how individual beliefs about uncertainty, metacognition, beliefs about beliefs, determine collective decision-making accuracy.

What if when you voted or shared your perspective on what we were going to have for dinner, you also conveyed an uncertainty estimate. How confident are you about that being the case? Finally, we show how agents can update their generative model over time, resulting in groups that are collectively more sensitive to external fluctuations and encode information more robustly.

From a distributed systems perspective, you can imagine two different kinds of distributed systems or networks. One of them has a dampening-like quality. So something that enters on one side dampens as it diffuses. We can also contrast that with another kind of distributed system or another kind of input or stimulus or context that amplifies. And the goal or one nice property to have a systems would be like, we want to dampen what we don't want to pay attention to and up-signal or increase the cascading availability of

what we do want to attend to. Now, that's a little circular since in some ways that's the definition of attention. However, making unified models like they do here gives us integrated approaches to explore a broad

range of collective phenomena here in spatial movement, but also involving the cognitive states of the particles. And they make a really nice representation here to connect to other approaches.

A traditional approach to classical self-propelled particles is agent environment. Agents take rules and they make decisions based upon those rules that are either given or learnt.

You know, when you're this close to the bird next to you, do that. And when you're this close, do this. And then social forces are these kind of as-if or actual constraints that then modify the agent's perspective. And what we see in the active inference setting is that the agent in the environment are mediated by an interface, the Markov blanket. This is the sensorium. This is on the inbound side, perception, and on the outbound side, action. And they give some more details. It's a really exciting paper. And it's just cool to see, again, that even in the preparatory days of a lecture like this, like there's always new collective activity in the ecosystem. And that's what makes it fun.

Last of the declarative sections of this lecture. Colonies here, but more generally, collectives can be understood as consisting of action perception loops which subsume, entail, or supervene upon individual level action perception loops. Niche modifications or stigmergy can also be modeled this way. What we did in this paper was bring something new to active inference in at least two different ways. First, it was the first model of an ant system of interest. Now, for some of us myrmecologists in the audience, all of our broader entomology colleagues, that may be enough. But what

was also really interesting about this work and the opportunity to collaborate on it was it was the first active inference model presented with explicit stigmergy, not just the philosophical consideration of the traces that agents leaving in the environment might then have them change their behavior in the future, but explicitly engaging with this process of multi-agent stigmatic behavior.

Active inference may be informative in the study of ant colonies at three distinct temporal scales.

Next, nest mate behavior, which is to say emergent among tissues of a nest mate,

you could keep on going collectively down, and interactions with the biotic or abiotic environment.

So first level, at the level of a nest mate, one of those things with six legs that looks like an insect.

At the level of the colony, colony development is emergent among nest mates and their interactions with the niche. That is where many people would traditionally assign collective behavior. They say, well, one nest mate walking around is individual behavior, that's the individual ants. And then the colony is where collective behavior comes into play, right? And we're going to see that reified very soon.

But lastly, we point to the evolution of populations, which is an emergent outcome amongst colonies within an ecology. In this work, we focused on the connection between the first two levels of analysis, focusing on the case of stigmatic colony foraging using a modification of previous active inference computational implementations. We didn't use the word individual, and there's a reason why. Because we're saying what we mean more clearly when talking about ants, when we talk about nest mates, colonies, and populations of colonies. That sidesteps or transcends the discussion of what is an individual and what is a collective. Because with respect to the colony as an individual, if you want to use that word, the population is a collective. With respect to the nest mate, the colony is a collective. With respect to the tissue, like the brain or the mid-gut, the nest mate is a collective. So we don't need to play the adjective game. We can just articulate what we're interested in about these systems of interest and complex adaptive system phenomena. And so what we did was we explored a kind of classical behavioral setting, which is food being placed between the arms of a teammates. And in this case, rather than a rat...

I'm just pausing for a second while I believe that my internet is reconnecting and we should be back.

Each ant was described by an action perception loop, an active inference action perception loop.

And they were able to move within a 3x3 grid, as well as perceive pheromone density within a 3x3 grid.

And then they laid down pheromone, they added stigmatic trace on the way home if they had found a seed.

And so this does a few interesting things and live stream 29 is where we unpack this a lot more,

but I just wanted to make two more points here. The first is that in a reward-centric perspective on ant behavior, the pheromone is seen as a semiotic cue and the seed or the food is seen as the reward.

And so then we got to use the cues, we got to use the pheromone density to like figure out where the good stuff is. That's that paradigm. We flipped that with active inference. Now the preference,

we don't need to say reward, but the preference, that which is sought after, proximally but teleologically, that which is sought after, preferred over is the pheromone. The seed becomes the semiotic cue

that it's time to turn on your pheromone depositing and come home. And so that flips

our approach to understanding what foraging is about from we're lost in the sauce, trying to figure out where the good thing is out there to we're surfing the hundred foot wave and we'll know it when we see

it and then we'll turn around. Very different perspective. And then one other point that I had to make for the context was that we used a modification of a previous active inference computational implementation. What was the paper that we drew from? In fact, it was the acquisition of culturally patterned attention styles under active inference with Axel and Alex et al. And what this paper did was they looked at the situation of visual foraging on a vase, a piece of pottery where the eye was only able to perceive a three by three matrix of pixels. And then the eye scanned around, looking to recognize broader patterns that were then culturally acquired. So maybe one culture or one subculture likes the checkerboard pattern and another one likes vertical lines. Another one likes horizontal lines or these other kinds of repetitive structures. What's so interesting is that the eye, when it is foraging, does not leave a trace, though it can feel like it, doesn't it? But the eye forages around in order to get the bigger picture based on local information. What we did is we said, well, instead of imagining that as the gaze location of an eye, let's imagine that as the skin in the game, exoskeleton on the ground of a forager. So let's think about that movement as something that an embodied forager does with its real body. And then conditional in this setting upon it touching the seed or the food, it can lay down a pheromone trail, which the eyes can't do, per se. And so just really interesting because it's not like this was an ant paper and now we're going to try to cram it into social sciences. It's actually like computational work in the social sciences was able to translate to mirror mycology into ants research. And so it's a very fruitful and bi-directional relationship between studying ants and studying people. Now we're going to move into the openness of questions in this second half of the lecture. So how did the concepts of generalized synchrony and homeostasis, which are core to active inference, relate to collective behavior? And here we can draw upon the work by Ralph Kaufman, Pranav Gupta and Jacob Taylor from 2021 called Inactive Inference Model of Collective Intelligence. And this was also unpacked in guest stream nine. And so here's a little collective over here. And that was about two years ago, way back when. What they did in this paper was they considered a setting with two trolley cars moving around a wheel. And so here's how they contextualized that work.

Formal models of collective intelligence have lacked a plausible mathematical description of the relationship between local scale interactions between autonomous subsystem components, individuals, and global scale behavior of the composite system, the collective. So they did a really great job faking out what they mean by individual and collective. In this paper, we use the Active Inference Formulation, AIF, a framework for explaining the behavior of any non-equilibrium steady-state system at any scale, to posit a minimal agent-based model that simulates the relationship between local individual level interaction and collective intelligence. Then they had a really creative approach to disentangling how different capacities translated to different collective outcomes. And here's what they did. We explore the effects of providing baseline active inference agents, model one, with specific cognitive capabilities. Model two added in a theory of mind. Model three added in goal alignment. And model four added both theory of mind and goal alignment. What does that look like? Again, we're not going to go into all the technical details here, but broadly this is what it looked like. Here was their simplest baseline active

inference formulation case. And then in the subsequent experiments, they added in either one or the other or both of the cognitive capacities that they described. They studied the single dyad case, and they very cleverly expanded that into a large-scale multi-dyad case. They utilized their own representation of the active inference partition in order to facilitate the construction of this model. And so in this case, we see the blanket with the sensory states coming in, the active states going out, Markov blanket.

Here we see the environment, and then internal belief states or cognitive states.

So, very cool work that helps us get a vocabulary around, not shut the book on, but focus on how do different agents come to synchronize? What is generalized synchrony? Well, it doesn't mean lockstep. Turn-taking is generalized synchrony because these two agents, the time that they're talking, has very high mutual information on the other. Even though the correlation with their speaking might be negative. Agent A is speaking when B is not, and vice versa. So, from a linear regression perspective, there's a negative correlation. But from a mutual information perspective, where you can't ever have below zero information, at worst, it's just noise. You always have zero or positive amount of information of systems about other systems. And so that opens the door to understanding generalized synchrony, which is a broader conception of how individuals in collectives are interacting. And homeostasis, maybe we could even imagine that these are multiple cells in the pancreas, and they're trying to coordinate on insulin sensitivity. Or maybe this is multiple people who are securing the defense of a community, and they also need to coordinate synchronously and asynchronously. And so there's a lot of ways in which this very, very lightweight apparatus of the active inference partition helps us bring together and compose all these cool what-ifs. Well, what if it had theory of mind? What if it had goal alignment? What if agents had theory of mind and goal alignment? What if we put them in a in a five by seven grid, and then we did this? And, and, and, and. That's what we get to do with active inference. And lastly here, how can active inference be used to explore the balance between individual and collective behavior and the bi-directional influence of collectives on individuals? And so here, I draw upon the amazing and singular work of Professor Mike Levin, who wrote *The Computational Boundary of a Self, Developmental Bioelectricity Drives Multicellularity and Scale-Free Cognition* in 2019. This was also explored in Livestream 25 series. So just to pull out a few of the great pieces of this article, and there's a stigmatic trail of many other papers for those who follow. Professor Levin writes, I propose a definition of an individual based on its information processing structure.

The scale and types of goals that a system can pursue defines or determines the boundaries and the content of the putative agent. On this view, what defines a coherent unified self out of its constituents components and the surrounding environment is the set of parts that operate towards reaching specific goal states. So this is a teleological or a functional or anticipatory or sentient or adaptive or informational concept on individuality, which we can add to the earlier consideration raised by Gilbert et al. Here they discuss some of the more mechanistic biological considerations like genetics, immunology, evolution, development, anatomy, and physiology.

What Levin has done here, and consistent with this is the informational theory of individuality by David Krakauer et al., is connect this more generally to system capacity. And so there is what is here

represented as like a time diamond where the scope or the scale of the system in the now is like the radius or the circumference on this diamond. And then the part of the diamond that recedes off into the past that's retained from the past is memory and history. And the part that advances or protends into the future is the anticipatory or the planning capacity. This opens the door to describing diverse intelligences, including possible alien artificial intelligence or synthetic intelligences, as well as nested or compound intelligences. So here the example is the ant colony. So here we have cells, these little pinkish diamonds nested within a nest mate, which is the green diamond nested within the colony, which is the blue diamond.

Now, notice that the nest mate is called an organism. I couldn't help but to connect this to Wheeler 1911, whose work is the ant colony as an organism. Superorganism came later. The superorganismal framing on the ant colony reifies the concept that the nest mate is the individual or the organism.

Because why else would you need to call the colony a superorganism? I mean, I think colonies are super. I think it's great to call them whatever you want to call them, and they probably don't mind either.

But the idea that a superorganism is composed of other organisms means that the nest mate is an organism. Whereas this diagram, in my perspective, would be as accurate or more to say nest mate here, because under different concepts of organismality, which is to say individuality, if we want to connect it there, the colony can be considered as the organism. And of the so many other pieces of this

work that are fun to bring out, the way that Levin clarifies his assumptions gives us, first off, some planks that we can jump off of for good and helps us come to terms with maybe like, how do we, what do we assume about behavior, about collective behavior? How do we want to show up when we're studying collective behavior in general? And when we're studying social behavior, after all, we're in active inference for the social sciences. And so Levin writes, the ideas are explored here from the perspective of the following three core assumptions. First is a commitment to evolution.

Every capacity has a natural history and emerged from simpler variants. Second, it is assumed that all metaphors are to be judged by their utility in driving scientific progress. There is not a binary categorization of scientific pictures, which should be taken literally or not, which can be decided a priori upfront. Third, much of the discussion centers around goals related to teleology, a hotly debated topic. Here, goal-directedness is taken in the non-magical, cybernetic engineering and control theory sense of a feedback system that operates to maximize some specific state of affairs, which can be modeled as a

dynamical system with attractors in its state space. And later work by Levin and Fields and others expanded this concept of intelligence just to phenomenal heights and flavors. So active inference helps us understand that cognitive glue. Maybe it's mechanistically enacted by bioelectricity in a given system or at a given scale, but active inference gives us the analytical glue to approach those systems with an open mind and an open scaffold. And so those are some of the insights and fundamentals of active inference as applied to collective behavioral systems. Now, on to the last third of this lecture, applications and implications of active inference for the social sciences regarding collective behavior. This is really where it is wide, wide, wide open.

What kinds of insights can active inference offer about the role of learning, adaptation, conflict, and cooperation in collective behavior? And how can we approach that from a cognitive security perspective? Here we'll draw on the work of Scott David and colleagues, Active Inference in Modeling Conflict. What implications does active inference have on the study of human collective behavior, particularly in social, economic, and digital contexts? Here we'll draw on the work by Poirier et al. Communication as socially extended active inference, an ecological approach to communicative behavior, Livestream 48 series.

What considerations are relevant for thinking about assemblages and interactions among different entities in cognitive ecosystems? Here we'll turn to the work of Friston et al. et al., 2022, designing ecosystems of intelligence from first principles.

And how does active inference accommodate collective behavior within and across generations?

And here we'll look at a variational synthesis of evolutionary and developmental dynamics by Friston et al. in 2023.

So, when it comes to applications and implications of active inference in social settings and settings, again, there's so much to do, there's so much to learn.

So, we welcome everybody who's curious. This is the time to write your live chats, submit your questions, prepare yourself to explore and to be an active forager in this space.

And if you're listening, you know that you already are.

So, in the work Active Inference in Modeling Conflict, Scott David, RJ Corday, and myself asked, what kinds of insights can active inference offer about the role of learning adaptation, conflict, and cooperation in collective behavior?

Now, as with all these other papers, there's always so many things that can be said.

What we did is we tried to connect beyond the very classical Markov blanket and agent environment partitioning,

which we've seen many, many, many times.

We tried to connect this to the specific kind of generative models that we use in modeling.

So, here we led early by clarifying, this is what we're talking about with active inference generative models.

It's not like something I'm hiding in my pocket and maybe I'll show you if you say the shibboleth.

This is what we're talking about when we're talking about applying active inference models to social settings.

It's a discrete time generative model here, just a motif.

We applied that kind of generative model architecture to one of the most well-known models of sense-making and decision-making,

which is the OODA loop, Observe, Orient, Decide, Act, by John Boyd.

Now, this may seem to some like a quite sparse framework, like, yeah, you get the observation, you figure it out, you make a decision, you take the action.

It doesn't do that much work by itself either.

However, OODA is used in tactical and strategic settings every single day.

And so, we wanted to connect OODA loop to the generative models that we use in active inference.

And so, we highlighted on the generative model where different stages of the OODA loop occur.

Perception happens in the sensory states.

Cognition, or the orientation, happens in the perceptual inference, the downstairs of that part of the figure.

And then, decision-making happens in the upstairs, with the expected free energy calculation in this case.

And finally, action is selected and enacted.

And that results in impact, and the cycle begins again.

Now, in active inference, it's more like an unfolding flow, but to a first-pass approximation, it does the trick.

Why does that really matter that we understand the mechanics of decision-making?

Well, we're talking about active inference in the social setting.

That means we're talking about two sophisticated cognitive entities, maybe two businesses, maybe two individual humans, maybe a person and a business, maybe different kinds of systems that we don't even know about or we haven't even seen yet.

And that generative model is the internal states that is then participating in this minimal dyadic collective behavioral encounter.

We can expand that to thinking about commons, resource commons, attention commons, open-source ecosystems.

And for this, we utilized one of Scott David's favorite acronyms, of which he has so, so many, which is BOLTS.

Business, Operation, Legal, Technical, and Social.

So, BOLTS is not an exhaustive acronym, otherwise we would need everything, but it's a heuristic that encourages us to think holistically about different systems.

Let's just say that we were a business or a person and we wanted to see if a given policy was a good way to go to complete our mission.

And it looks good, except it can't be done operationally.

Well, then that has to be taken into account.

Or everything works out, but it's illegal.

That's not the way to go.

Everything works out, but the business doesn't check out.

It's not going to be sustainable financially.

Or technically, it can't be done.

Or socially, it's not palatable.

All of these features of socio-technical systems, which again, we just make a little mnemonic of with BOLTS, need to be taken into account for comprehensive and high-integrity, high-reliability systems design.

In this setting, we have Entity A and Entity B interacting through the BOLTS niche.

They each have their own blanket states, sense and action.

And the niche, where those nestmates were laying their pheromones, the niche is BOLTS.

And so you could make a restricted model where it's just two entities engaged in a financial dealing, just a business, only engaging on B.

Or you could think of this multi-domain, full-spectrum situation in terms of BOLTS.

Then you would have a larger and a more complex interface between these two entities.

But we also wanted to think about the case where there might be cooperation between these entities, or there might be conflict.

And so here we have Entity A and B.

The dyad is in the spotlight.

Entity C is a non-belligerent state.

Or it's a bystander.

Or it's a witness.

Or it's a referee.

Or it's a friend.

Or it's a colleague.

But Entity C is also engaging only through their blanket states on BOLTS.

And lastly, we connect it to the abiotic niche, which may not care how much you're buying or selling a gallon of oil for.

The abiotic niche may have its own dynamics.

We may use active inference to model that, or we might use some other approach to model it.

And so this sets a really, really general state space for us to understand conflict and cooperation.

And where we situated that work, lastly here, was in relationship to other work exploring strategic and tactical decision-making.

And so on the lower end of this image are tactical and operational frameworks that are used in decision-making.

On the top are strategic or scholarly frameworks that are used for more of geopolitical considerations.

And then from left to right, we have more qualitative approaches going to more formal and quantitative approaches.

And we believe that the application of active inference through this type of model, the active inference conflict model,

helps us touch the qualitative, natural language, and graphical, composable aspects, and the quantitative and formal aspects as well.

And similarly, through nested decision-making cycles, we can talk about tactical operational activities and the broader nested strategy,

just like we looked at the nested cones from Professor Levin's work.

Next, what implications does active inference have on the study of human collective behavior, particularly in social, economic, and digital contexts?

Well, there's a lot to say about this, but Tisson and Poirier wrote in 2021 the paper,

Communication as Socially Extended Active Inference, an Ecological Approach to Communicative Behavior.

And this was in the Livestream 48 series.

Here's Remy raising his hand so that we knew that he wanted to speak.

They introduced an ecological account of communication according to which

acts of communication are active inference achieved by affecting the behavior of a target organism via the modification of its field of affordances.

Constraining a target organism's behavior constitutes a mechanism of socially extended active inference, allowing organisms to proactively regulate their inner states through the behavior of other organisms.

In this general conception of communication, the type of cooperative communication characteristic of human communicative interaction is a way of constraining interaction dynamics towards the goals of a given joint action by constructing and altering shared fields of affordances.

This account embraces a pragmatist view according to which communication is a form of action aiming to influence the behavior of a target and stands against the traditional transmission view according to which communication fundamentally serves to convey information.

Understanding acts of communication as active inference under an ecological interpretation allows us to link communicative and ultimately linguistic behavior to the biological imperative of minimizing free energy and to emphasize the action-oriented nature of communicative interaction.

This is really a rich and excellent work that Remy and Pierre produced.

And there are several pieces that I think add a lot of meaning and value to the ways that we think about communication.

But most broadly, what is communication for?

Who do we talk to and why?

What do we like to talk about and why?

Why would anyone say that and why?

How does technique, considered broadly, technology and technique, whether it's digital or social or cultural technique, mediate and influence communication?

The media is the message.

The medium is the message.

And think about everything in that area too.

And how can active inference, beyond just giving these incredible, composable, formal tools, how can we think about joint action and co-modification of fields of affordances?

Hey, could you pick that up from that shelf?

Okay, I'll open it for you.

Now let's hand it to our friend who's going to make the food.

All of these ways in which individuals shape each other's field of affordances is something that we can talk about really fluidly within active inference.

Penultimate view.

What considerations are relevant for thinking about assemblages and interactions among different entities in cognitive ecosystems?

So here was a large multi-author paper from Carl Friston et al.

and the Versus colleagues,

Designing Ecosystems of Intelligence from First Principles from December 2022.

And the big picture as I see it here is,
what is the role of the social sciences,
after all, the course that we're in now,
in ecosystems of diverse, heterogeneous, synthetic intelligences?
By synthetic intelligences, I refer to assemblages with intelligence.
Whether it's natural or artificial, leave that debate for another day.
Because in their application, we're talking about synthetic intelligence.
Synthetic intelligence needs cognitive security.
Cognitive security needs cognitive science.
And they began this paper with the following.
This white paper presents active inference as an approach to research and development
in the field of artificial intelligence, AI.
Artificial.
With the aim of developing ecosystems of natural and artificial intelligences.
The path forward in AI is often presented as progressing from systems
that are able to solve problems within one narrow domain,
artificial narrow intelligence, ANI,
to systems that are able to solve problems in a domain general manner,
at or beyond human levels,
what are known as artificial general intelligence, AGI,
and artificial superintelligence, ASI.
We believe that approaching ASI, or for reasons outlined below, even AGI,
likely requires an understanding of networked or collective intelligence.
Now, intelligence in action is behavior.
So we're talking about collective behavior.
And they laid out their timeline and the stages that they see
in terms of sentient, sophisticated, sympathetic, and finally shared intelligences.
But I think it's worth to dwell a moment on the one figure of this paper.
Figure one.
Belief updating on a statistical manifold.
And so here we have some curve in an information geometric space.
It's a parameter space of a probability distribution.
And on this manifold, on this line on a plane,
lower dimensional manifold projection,
there's a temporal process, belief at time t , t plus 1, t plus 2,
where initially this Gaussian that's very distributed
becomes updated to a sharper peak distribution
that's going to be resting basically at some special point on this curve.

Whether you're very familiar with accelerated optimization on information geometric manifolds or not, it's really interesting to consider why they went with this image. And for those, again, who are familiar or just coming to be familiar with some of the formalisms of active inference, this is what this very talented and transdisciplinary team went with when seeking to provide some of the only non-textual descriptions of their work. So I think it's really interesting.

And the big question is, well, what is the social?
Reassembling the social.

Bruno Latour.

What is the social?

What is social?

Does social mean only adult humans talking with each other?

Does social include non-human species?

Does social include non-biological intelligences?

These are all things that are not just questions of theory.

And so connecting flexible, extensible frameworks like active inference and all that comes along with it to this bigger question about applications and implications of applied active inference in the social setting. That's what I hope we can explore in the questions at the end of this live stream and in the discussions that we have in a couple of weeks and ongoing. And lastly, how does active inference accommodate collective behavior within and across generations?

And so here I'll draw on the recent work of variational synthesis of evolutionary and developmental dynamics with Carl Fristen, myself, and others. And we had two images in this work, which otherwise had a lot of prose and formalism. On the left, we see that active inference particular partition, external states, internal states, blanket states.

Here at the organismal level, one worm, and in an innested, deeper system, the population, which gives rise to the starting positionality of the organisms.

Think about drawing an organism from a gene pool, from a birth pool, and the impact that those organisms have on the bigger picture.

Maybe it's a bacteria that is fermenting.

So it changes the pH or the ethanol concentration of the kombucha.

That is this kind of multi-scale interaction that we believe provides an approach to understanding eco, evo, devo, ecology, evolution, and developmental interactions.

And the other visualization that we use here is of nested systems.

So contrast that with the kind of time diamond that we saw from Mike Levin.
Now here, imagine that we're going to cut across the center of this diamond.
And now it's going to look like a muscle fiber, fibers of fibers.
Because we have, let's just say, this outer blue line may be the colony.
And then each of these smaller entities is an estimate.
And they're composed of cells.
And cells have constituents too.
And here, we utilized the apparatus of the renormalization group, which draws upon physics,
to articulate the separation of scales so that we can be clear.
When are we talking about multi-scale systems, which we can articulate and renormalize across vertically,
to use an encultured metaphor.
And then within a given scale of analysis, what is the lateral or collective behavior?
So how do we get those interactions amongst nestmates that are both constrained by the colony,
top down, again, using a spatial and encultured metaphor,
top down, scaffolded or enabled or constrained by the colony.
But then what nestmates do matters.
And that influences the colony.
And we used a few new pieces of formalism, primarily developed by Carl,
to help express that setting.
And again, we think it's going to be really useful for describing
otherwise quite challenging scenarios in ecology, evolution and development.
And so we come right back to that.
What is a self?
What about nested selves?
And with that, I turn to some open questions.
What can be said other than?
There are so many open questions to discuss.
There's many things we can explore in the discussion that I'll be facilitating in a couple weeks.
And of course, beyond.
We have a CODA interface for you to submit questions that all the course teachers will take a look at
and give their best attempt on.
And conversation is collective behavior.
Now, sure, keeping it inside of your head is also collective behavior,
no matter how you think about it.
But when we enact in the social, then we're really making it happen.
So with that, I will just show my citations.
And the slides will be shared too.
Also, there's some extra readings that various individuals along the way suggested to me.

So that includes social ecologies and abductive logic, enacting like a state.

These are not the titles.

The titles are inactive dynamic social cognition and active inference.

That was live stream 47.

Thinking like a state embodied intelligence in the deep history of our collective minds.

That was live stream 33.

Active inference applicability to different kinds of social organizations explained through reference to industrial engineering and quality management.

That was live stream 27.

Resonating minds.

Emergent collaboration through hierarchical active inference.

Yella Broomberg's and Eric Raittvel's work on self-organization, free energy minimization, and optimal grip on a field of affordances.

And the well-known thinking through other minds work by Vessier et al., thinking through other minds, a variational approach to cognition and culture.

Well, I hope this has been an interesting and relevant lecture for you.

It was fun to prepare.

There are so many perspectives and so much territory that there's a lot in collective behavior.

But again, I'm just really looking forward to the discussion that we can have in a couple of weeks.

I encourage you, whether you're listening on your timeline, at a moment that enables you to join that live conversation, I would encourage you to join.

Or to join the live stream so you can participate through chat.

Or whether you want to find some other way to get involved.

It's been a great time.

And with that, I will close.

Thank you very much.

Until next time.

Bye.